Task A: Tutor hiring funnel review

The September snapshot shows lower signup-to-induction conversion than August: 155 of 3,377 signups (4.6%) compared with 208 of 2,602 (8.0%). September had less time to complete the funnel by the 5 October data cutoff, so I would treat that comparison as provisional.

The clearest immediate signal is on mobile. September v2 mobile applicants passed the AI test at 11.7%, compared with 43.5% on v1 mobile. Desktop rates were close (48.5% on v2 and 46.1% on v1). I would check the mobile test path and version assignment, then run a controlled v1/v2 comparison on mobile before making a wider change.

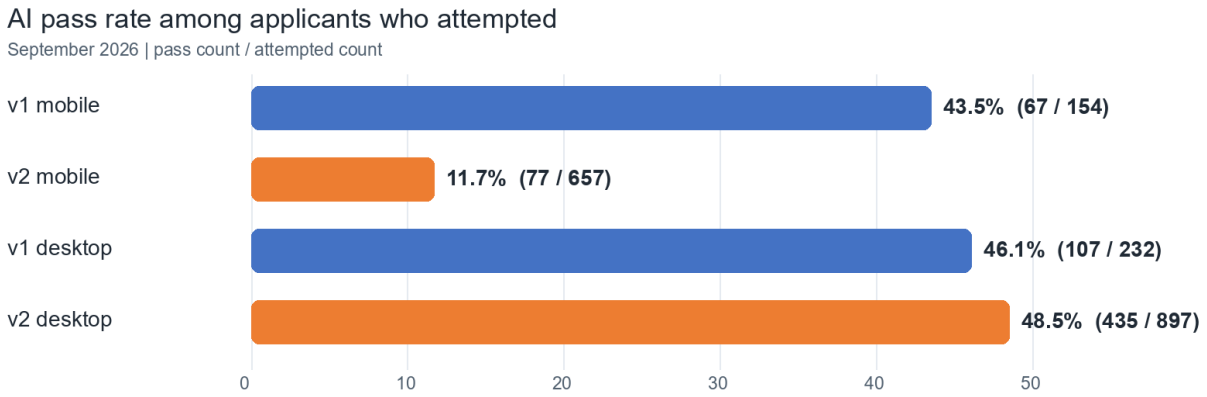


Figure 1. Rates use AI passes divided by test attempts.

Where the overall rate changed

Funnel step	August	September	Contribution
AI attempted / signups	63.5%	57.4%	−0.62 pp
AI passed / attempts	46.2%	35.4%	−1.64 pp
CET / AI passers	50.1%	54.4%	+0.51 pp
Interview / CET clear	84.9%	80.4%	−0.33 pp
Interview passed / interview	73.5%	62.0%	−1.05 pp
Induction / interview passed	87.0%	83.3%	−0.27 pp

The symmetric six-step rate decomposition adds to the 3.40 percentage-point decline. The largest negative contributions are the AI pass rate and interview pass rate; CET conversion improved and partly offset the decline. These are descriptive contributions, not proof of cause.

A scale check gives the mobile issue some context: applying the v1 mobile pass rate to the 657 v2 mobile attempts implies about 209 additional passes. If those applicants converted downstream at September's observed rate, that would mean roughly 47 more inductions, or about 1.4 points of the gap. This is only an estimate and is not added to the funnel decomposition.

Acquisition mix also shifted: Meta rose from 25.5% to 40.3% of signups, while Referral fell from 20.3% to 13.5%. A descriptive source mix/rate split attributes about −0.95 points to mix and −2.46 points to within-source rate changes. Only 573 signups from 1–5 September had a full 30-day follow-up window by the cutoff; 37 attended (6.5%). I would compare fixed-age cohorts before treating September's 4.6% as final.

Task B: Repair the weekly report

I corrected three issues that affected the weekly report: a source label could stop it, the AI pass rate used the wrong denominator, and the report did not show the requested CET rate.

Issue	Change	Result
Source label crash	Normalize case, map facebook to Meta, and keep unknown or blank labels visible.	New source values no longer stop the report.
AI pass denominator	Divide passes by applicants who attempted the test, not by all signups.	Aug: 764/1,652 = 46.2%; Sep: 686/1,940 = 35.4%.
CET rate missing	Add CET passes divided by AI passers.	Aug: 383/764 = 50.1%; Sep: 373/686 = 54.4%.

How I checked it

I ran the report on the supplied candidate CSV and checked the month totals, source mapping, AI test counts, and CET column. Four standard-library tests cover source aliases and unknown values, both rate denominators, and the agent's sender-ID lookup guard. All four passed. The report still uses the same weekly grouping and command-line workflow, and it needs no third-party Python packages.

Why the denominator matters

Dividing AI passes by all signups gives 29.4% for August and 20.3% for September. That measures passes as a share of signups, not the test team's pass rate among people who took the test. Using attempts as the denominator gives 46.2% and 35.4%, which is the measure that matches the roughly 40% reference in the ticket.

For reporting, facebook is treated as another label for Meta; the original source value remains unchanged in the data. Unknown values are retained as their own groups so they can be reviewed instead of silently discarded.

Task C: Applicant email agent

I designed the agent to prepare a reply decision for human review. It does not send email or change candidate records. The model can request a limited lookup, while local code enforces which candidate record it may see.

How a message is handled

Step	What happens
1. Read	Load one inbox row and its received date.
2. Check	The model decides whether it needs the lookup tool. Local code returns status fields only for the candidate ID on that message.
3. Decide	Apply the policy and choose a draft reply or escalation.
4. Validate	Check the action and required fields before saving.
5. Save	Checkpoint the draft decision so a person can review it and the run can resume.

Live test and access limits

The live run processed 0 of 15 messages. An API key was configured, but the first request failed during DNS/network connection before OpenAI returned a response. That means I could not measure the account's rate limit or quota. ChatGPT subscription access and API usage are separate.

I manually checked the 15 expected replies against the policy and candidate records, including retake, reapplication, and induction-date cases. These replies are a reference baseline, not model output. The model-run checkpoint is empty because no request completed. The code can be rerun with a small message limit once the API endpoint is reachable; completed message IDs are saved so the run can resume.

Safety, likely errors, and cost

Email text is treated as untrusted input. The lookup is read-only and restricted in code to the sender's candidate ID, so a request for another applicant's details cannot retrieve that record. The model sees only the status fields needed for the response. Complaints, ambiguous messages, unsupported questions, requests to change results or personal data, and staff claims are routed for review. Drafts should be checked before anyone sends them.

A rough estimate is about $5 per 1,000 emails on GPT-5.4 Mini, assuming 5,000 input tokens and 300 output tokens per email, including the tool-call turn. Actual cost depends on prompt length, tool calls, and current prices. The estimate uses published rates of $0.75 per million input tokens and $4.50 per million output tokens. No account-specific usage estimate is available because the request did not reach the API.

TASK D: November operating plan

The target is 75 induction attendees each week for four weeks, or 300 in total. Using September's 4.59% signup-to-induction snapshot as a planning rate means about 1,634 signups per week (6,536 over four weeks). At September's 51.7% interview-to-induction rate, the team needs roughly 145 interview slots each week. These are planning estimates, not guarantees; signups need to start early enough for interviews and Monday inductions.

Interview capacity

Current capacity	Estimated need	Shortfall	Plan
12 interviewers × 9 = 108 slots/week	~145 slots/week	~37 slots/week (34%)	Add 5 trained temporary interviewers: 17 × 9 = 153 slots/week.

I would add five trained temporary interviewers. This closes the estimated gap with a small buffer, though onboarding and calibration take time. More paid blocks for current staff could be faster, but would exceed the nine-interview cap. Improving earlier funnel conversion and smoothing schedules is useful over time, but is less certain as an immediate fix.

Monday checks and escalation points

Check	Act when	Owner
Signups vs plan	Projected below 1,634 this week	Marketing lead
AI attempt rate	Below 57%	Hiring operations
AI pass / attempts	Below 35%	AI test owner
v2 mobile pass (n ≥ 50)	Below 35% or >10 pp below desktop	Test/product owner
CET / AI passers	Below 54%	CET owner
Interview slots	Need more than 108 confirmed	Supply lead
Interview pass rate	Below 62% on a mature cohort	Interview lead
Induction attendance	Forecast below 75 next Monday	Induction coordinator
Source quality	Any unmapped source or >1% blank	Data owner

Work estimate and correction

Estimated effort: Part A 75 minutes, B 30, C 100, D 45, and packaging/checks 30, for about 4 hours 40 minutes. This is an estimate, not a time log. AI assisted analysis and code drafting; I checked the numbers against the CSV and independent calculations. One early calculation treated AI passes as a share of signups. I corrected it to passes divided by attempts, which matches the requested AI test pass rate.